

Zewail City of Science and Technology
University of Science and Technology
Computational Sciences and Artificial Intelligence (CSAI) Program
DSAI 201, Data Mining and Information Retrieval, 3 Cr, Spring 2026, Required
Assoc. Prof. Mohamed Maher Ata – momaher@zewailcity.edu.eg

Weeks	Lecture	Assessments and laps
Week 1	- Introduction to data retrieval - Boolean Retrieval	
Week 2	Pre-processing for data retrieval	
Week 3	Inverted index	Start course Project: Search engine design
Week 4	- Evaluating Search Effectiveness - Ranked Retrieval I: Vector Space Model - BM25 Model	Quiz 1
Week 5	- Ranked retrieval - TF-IDF	
Week 6	- BM25 OKAPI ranking function - Vector space model	Lab Exam 1
Week 7	- Probability ranking principle - Language models - Jelinek mercer smoothing	Project discussion phase (1)
Week 8	Midterm	
Week 9	- Query Expansion - Relevance Feedback - ROCCHIO algorithm	
Week 10	Sham El-Nessim (Spring Holiday)	
Week 11	- Local term Representation - Distributed term representation - Word embeddings	Quiz 2
Week 12	Main concept of RNN and EIMOs	Lab Exam 2
Week 13	Ranked Retrieval III: Neural Models (transformers)	
Week 14	Attention and positional encoding	Project discussion phase (2) Posting the final course work
Week 15	Bidirectional Encoder Representations from Transformers (BERT) for Re-Ranking	
Week 16	Final Exam	

Grading system

	Total number	Grade of each topic	Total weight
Lap attendance and practice	10	1	10
Quizzes	2	5	10
Lap Exam	2	5	10
Project	1	10	10
Total weight			40
Midterm			20
Final term			40

Important notifications for the course policies:

- To pass the course, a student must achieve at least **35%** in the final assessment.
- A minimum of **75%** attendance in both labs and lectures is required for students to pass the course.
- Any medical excuse must be approved by the **UST clinic**, and the student is responsible for forwarding the clinic's approval to the assigned instructor and teaching assistants within **two** working days. Failure to do so will result in the student being held accountable for any delays, and makeup opportunities may be disregarded.
- The use of AI tools to complete any assessment is strictly prohibited. Any student found violating this policy will receive a **zero** for the affected assignment and the final course work. This policy is in place to ensure that each student gains a comprehensive understanding of the subject matter through their own efforts and learning. Please adhere to this policy to maintain academic integrity and fair assessment.
- Please be aware that the attached detailed syllabus content may undergo some modifications throughout the semester. Topics may be added or removed based on the instructor's vision, the collective understanding of the class, and the available time within the semester. Your feedback and engagement will also guide these adjustments to ensure the course meets both learning objectives and time constraints.
- For any questions or clarifications, students are encouraged to reach out during office hours or via official email. Please allow up to 48 hours for a response, especially for inquiries sent outside of office hours. Please note that communication via WhatsApp or phone calls will not be accepted. Official office hours will be announced soon on the classroom platform.

Course Project: Design and implement a basic search engine for a collection of text documents.

Project Procedures:

Phase (1): 4 Points

1. Data Collection: (1 Point)

- Gather a set of text documents to serve as your corpus. This could be a collection of articles, web pages, or any other textual content.
- Ensure the documents are in a format that can be easily parsed and indexed
- Here are **some options** for the process of gathering **a set of text documents** to serve as a corpus for your project:

i. Sources for Text Documents:

1. **Online Repositories:** Platforms like Kaggle, GitHub, Hugging Face, UCI Machine Learning Repository, and others offer a wide range of text datasets covering various domains. **(Recommended)**
2. **Test Collections:** any test collection datasets such as **TREC** (the Vaswasni dataset) which contain a built-in (topics file, qrels file....)
3. **Web Scraping:** You can scrape text data from websites using libraries like BeautifulSoup or Scrapy.
4. **APIs:** Some websites provide APIs to access their textual content, which can be utilized to gather data.
5. **Publicly available datasets:** Many organizations and research institutions share their datasets for public use.

ii. Document Formats:

- Choose a format that can be easily parsed and indexed for text analysis. Common formats include plain text files (**.txt**), **CSV files**, **JSON files**, or **structured databases**.

iii. Building the Corpus:

- Combine the **text documents into a corpus**

2. Preprocessing: (1 Points)

- Tokenization: Split documents into individual words or tokens.
- Lowercasing: Convert all text to lowercase for case insensitivity.
- Stopword Removal: Eliminate common words (e.g., "and", "the", "is") that do not contribute much to the meaning of the document.
- Stemming or Lemmatization: Reduce words to their base or root form (e.g., "running" to "run").

3. Indexing: (2 Points)

- Build an inverted index
- Create a data structure that maps each unique word (or term) to the documents that contain that word.
- For each term, maintain a list of document IDs where the term appears along with the frequency of occurrence.

Phase (2): 6 Points

4. Query Processing: (2 Points)

- Implement a simple query processing with expanded capabilities
- Parse user queries (input text) and apply the same preprocessing steps (tokenization, lowercase, etc.) as used during indexing.
- Identify relevant documents by leveraging the inverted index
- Retrieve documents that contain all the terms from the query.
- Rank the retrieved documents based ranking algorithm (TF-IDF).

5. Query expansion: (2 Points)

- Apply relevance feedback by analyzing top-ranked documents for initial queries.
- Incorporate synonyms or related terms using pre-built mappings or **Embeddings (RNN, ELMo and BERT)**.

6. User Interface: (1 Point)

- Develop a basic user interface to interact with the search engine
- Accept user queries.
- Display relevant search results.

7. Evaluation: (1 Point)

- Evaluate the performance of your search engine:
- Test with various queries to assess retrieval accuracy and speed.

For the final phase:

Project submission requirements:

1. Source Code:

Upload the complete source code of your search engine implementation via classroom before the final discussion session, with a deadline of within 1 complete day (**24 hours**) prior to the discussion. This deadline ensures that all submissions can be reviewed prior to our final discussion and allows time for any necessary preparations.

2. PDF Report:

Prepare a PDF report that contains a brief description of each stage of your search engine implementation. Each student will upload his/her PDF at the same time of uploading the source code.

3. Bonus

A bonus may be awarded based on the quality of the project's final outcome. The bonus is reserved for **extraordinary work** in terms of design quality, search engine performance, and implemented features. The Teaching Assistant (TA) will evaluate the project and determine whether a bonus is granted. Bonus marks range from **1 to 5 points**, depending on the overall design, search engine capabilities, and user interface quality.

Important precaution:

As a reminder, **the use of AI tools** to complete this project is strictly prohibited. Any student found violating this policy will receive **a zero** for the affected assignment and the final **course work**. This policy is in place to ensure that each student gains a comprehensive understanding of the subject matter through their own efforts and learning. Please adhere to this policy to maintain academic integrity and fair assessment.